Protocol

Cat. No. K4207

Anti-DYKDDDDK Magnetic Beads

Introduction

Flag Tag is a polypeptide composed of 8 amino acid residues (DYKDDDDK), typically existing in the form of Flag or 3×Flag. It can be fused to the 5' or 3' end of a target gene via genetic recombination technology, resulting in a target protein with a Flag tag. The Flag tag is widely used in various research areas such as protein expression, purification, identification, interaction, and functional studies due to its characteristics: it does not interact with the target protein and does not affect the protein's function; the N-terminal Flag tag can be cleaved by enterokinase to obtain a tag-free target protein; and it can be detected and purified using Flag antibodies, Anti-Flag magnetic beads, or Anti-Flag affinity gel.

Anti-DYKDDDDK Magnetic Beads are a biological research tool specifically designed for life science research—Flag tag immunomagnetic beads. They are produced by covalently conjugating high-quality Flag monoclonal antibodies to nano-scale amino magnetic beads. The Flag antibodies are immobilized on the surface of the magnetic beads, enabling specific binding to proteins containing the Flag tag. Subsequently, target proteins can be rapidly purified and detected through magnetic separation technology.

This product can be used for applications such as immunoprecipitation (IP), co-immunoprecipitation (Co-IP), protein purification, protein expression, and functional studies. It is available in two specifications: 1 mL and 2 mL. Taking a single sample (e.g., 10^5 cells) requiring 25 μL of magnetic beads as an example, the 1 mL specification can be used for 40 samples.

The product parameters are as follows:

Product name	Anti-DYKDDDDK Magnetic Beads
Product Size	1 mL / 2 X1 mL
Matrix spherical	Nanoscale amino magnetic beads
Ligand	Anti-Flag Tag monoclonal antibody (equivalent to Sigma M2 antibody).
Particle size	1 μm
Binding Capacity	≥ 0.6 mg DYKDDDDK-tagged fusion protein/mL of beads.
Concentration	10 mg/mL
Storage Buffer	1X PBS, 0.1%BSA, 0.01% Tween-20, 0.05% Proclin-300.
Application	IP, Co-IP, and Purification of small-scale DYKDDDDK (Flag) - tagged proteins
Storage	Store at 4°C for 12 months, avoid lower temperature storage

Protocol

1. Reagents to Be Prepared by the User

- Binding/Wash Buffer: 50 mM Tris, 150 mM NaCl, 0.1%–0.5% detergent (Triton X-100, Tween 20, or NP-40), pH 7.5.
- Elution Buffer (choose one of the following):
  a. Competitive Elution: It is recommended to use 3X FLAG® Peptide (Cat. #: A6001) for higher elution efficiency.
  b. SDS-PAGE Loading Buffer Elution: It is recommended to use 5X Protein Loading Buffer (Reducing) (Cat. #: K1164) for subsequent electrophoretic analysis.
  c. Acidic Elution: It is recommended to prepare a solution of 0.1 M–0.2 M Glycine, 0.1%–0.5% Detergent, pH 2.5–3.1 (or 0.1 M citric acid, 0.1%–0.5% detergent, pH 2.5–3.1).
- Neutralization buffer (for acid elution): 1 M Tris, pH 8.0.

2. Sample Preparation

- Adherent cells: Remove medium and wash twice with 1× PBS (Cat. #: H2018) at a ratio of 150 μL per 1.0×10^5 cells; Scrape off the cells with a cell scraper (or collect the cells after normal digestion), collect them into a 1.5 mL centrifuge tube, add binding/wash buffer at a ratio of 20~30 μL per 1.0×10^5 cells, add protease inhibitors, mix well, and incubate on ice for 10 minutes. Centrifuge to collect the supernatant (14,000 × g at 4 °C for 10 min) and set aside on ice (or store at -20 °C for a long time).
- Suspension cells: Centrifuge to collect cells (centrifuge at 500×g at 4°C for 10 min), discard the supernatant, weigh it, and wash 2 times in 1× PBS (Cat. #: H2018) at a rate of 50 μL per mg. Binding/wash buffer was added at a ratio of 5~10 μL per mg of cells, and protease inhibitor was added, mixed and placed on ice for 10 min. Centrifuge to collect the supernatant (14,000 × g for 10 minutes at 4°C) and set aside on ice (or store at -20°C for a long time).
- Escherichia coli: Centrifuge collect Escherichia coli (centrifuge at 12,000×g at 4°C for 2 min), discard the supernatant, weigh it, and wash it twice with 1×PBS at a rate of 10 mL per gram of bacteria (wet weight); Binding/wash baffle was added at a ratio of 5~10 mL per gram of bacterial body (wet weight), protease inhibitor was added, bacterial body was resuspended, cells were lysed on ultrasound, and supernatant was collected by centrifugation (12,000×g centrifuged at 4°C for 10 min).

***Note:**

1. It is recommended to use Cell Lysis Buffer for WB and IP without Inhibitors (Cat. #: K1124) as the lysis buffer. It is a highly efficient lysis buffer for tissues or cells.
2. Recommendation for Protease Inhibitor: For prokaryotic expression systems, it is recommended to use Protease Inhibitor Cocktail (EDTA-Free, 100X in DMSO) (Cat. #: K1007).

3. Magnetic bead pretreatment

Since Anti-DYKDDDDK Magnetic Beads are stored in a special protective solution, they need to be properly washed before adding to the sample.

a Gently resuspend the Anti-DYKDDDDK Magnetic Beads by pipetting. Transfer 25 μL of the bead suspension to a clean 1.5 mL centrifuge tube.

b Add 500 μL of binding/wash buffer, gently blow with a pipette and fully resuspend the Anti-DYKDDDDK Magnetic Beads. Place on a magnetic rack for 1 min to remove the supernatant.

c Repeat step (3.b) 1-2 times above.

4. Immunoprecipitation, IP

a. Add 500 μL of the sample prepared in step 2 to the pretreated magnetic beads and incubate on a flip mixer (1 hour at room temperature; or 4-6 hours/overnight at 4°C).

b. Place the mixture on a magnetic stand for 1 minute. Transfer the supernatant to a new centrifuge tube (the supernatant can be used to detect any residual DYKDDDDK-tagged protein). The remaining material in the original tube is the protein-bead complex.

5. Washing

a Add 500 μL of Binding/Wash Buffer to the complex from the previous step (4.b). Gently resuspend by pipetting, then place the tube on a magnetic stand for 1 minute and remove the supernatant.

b Repeat the step 3 times until the washed supernatant OD280 is less than 0.05.

***Note:** If the supernatant OD280 is greater than 0.05, the number of washes should be increased appropriately.

6. Elution

It is recommended to choose one of the following three elution methods based on the characteristics of the tagged protein and the requirements of downstream experiments.

■ 3X Flag competitive elution method: This method is a native method, which has high elution efficiency and maintains the original biological activity of the eluted protein, which is convenient for subsequent analysis and detection.

a Preparation of 3X Flag peptide eluent: Dissolve an appropriate amount of 3X Flag peptide (A6001) in 1X TBS at a final concentration of 150 μg/mL.

b Add 100 μL of 3X Flag peptide eluent (150 μg/mL) per 10-20 μL of the original bead volume, mix well, and place on a side-swing shaker or rotary mixer and incubate at room temperature for 30-60 min, or 4°C for 1-2 h. To improve elution efficiency, incubation time can be extended or repeated elution can be done.

c After incubation, separate on a magnetic rack for 1 min and transfer the supernatant to a new centrifuge tube. The supernatant is the eluting Flag-tagged protein.

d The eluted Flag-tagged proteins are set aside at 4 °C or stored at -20°C or -80°C for extended periods.

- Acid elution method: This method is a native method, which is fast and efficient. The eluted protein retains its original biological activity and facilitates subsequent functional analysis.

  a Add 50-100 μL of acidic eluent to the washed protein-bead complex species and incubate at room temperature for 5-10 min, but not more than 15 min.

  b After incubation, the centrifuge tubes were placed on a magnetic rack for 1 min to separate and the supernatant was transferred to a new centrifuge tube.

  c Add 10 μL of neutralization bbuffer per 100 μL of eluent to adjust the pH of the elution product to neutral for later functional analysis.

  d Eluted and neutralized Flag-tagged proteins are set aside at 4°C or stored at -20°C or -80°C for extended periods.

***Note:**

1. Acid elution, although highly efficient, may still be inferior to competitive elution methods or SDS-PAGE loading buffer elution.
2. Since the difference in the target protein may have a certain impact on the elution efficiency of the acid elution method, if the requirements for elution efficiency are relatively high, the pH of the acid eluent can be adjusted between 2.5-3.1, and the pH value or amount of the corresponding neutralization solution should also be adjusted to a certain extent.

- SDS-PAGE loading buffer elution method: This method is a denaturation method, and the protein samples obtained are suitable for SDS-PAGE electrophoresis or WB detection.

  a Preparation of SDS-PAGE loading buffer: Dilute the prepared SDS-PAGE loading buffer with water to a 1× concentration. Note that the loading buffer contains reducing agents such as DTT, and the eluted protein sample will include the light and heavy chains of the Flag antibody.

  b Add 80-100 μL of 1X SDS-PAGE loading buffer to the washed protein-bead complex species and heat at 100°C for 10 min.

  c After cooling, it was placed on a magnetic rack and left for 1 minute, and after the magnetic beads were adsorbed on the side wall of the centrifuge tube, the supernatant was collected for SDS-PAGE electrophoresis or Western blot detection.

FAQs

1. How to avoid magnetic bead aggregation?

   Answer: You can choose the following methods: 1. Storage conditions: Magnetic beads should be stored in an environment of 2~8°C; 2. Prevent contamination and dryness: Avoid contamination during use and ensure that the beads do not dry out; 3. Add detergent: Adding 0.1% non-ionic detergent (such as Triton X-100, Tween-20, or NP-40) to the binding/wash buffer can effectively prevent magnetic bead aggregation.

2. How to solve the problem of magnetic beads adhering to the tube wall?

   Answer: 1. Use low adsorption consumables: When performing magnetic bead operation, it is recommended to use experimental consumables with low adsorption rate; 2. Add detergent: Add 0.01%~0.1% non-ionic detergent to the buffer to reduce bead adhesion.

3. No target protein detected after IP?

   Answer: 1. Insufficient antigen content: ensure that the antigen content in the sample is sufficient and can be verified by SDS-PAGE or Western Blot; 2. Increase sample dosage: If necessary, increase sample dosage; 3. Buffer Component Interference: Try using other buffers for immunoprecipitation and rinsing.

4. How to solve the problem of low target protein amount?

   Answer: 1. Protein degradation: add protease inhibitors to prevent protein degradation; 2. Increase the amount of magnetic beads: If the amount of Anti-DYKDDDDK immune magnetic beads used is insufficient, consider increasing the dosage.

5. What causes non-specific bands to detect?

   A: Non-specific protein binding: Add 50~350 mM NaCl to the binding/wash buffer and increase the number and intensity of elution.

Note

1. This product needs to maintain a pH of 6-8 to avoid high-speed centrifugation and drying; Do not place the beads in the magnetic field for a long time, otherwise it may cause the beads to agglomerate.
2. Before using this product, it should be properly and fully resuspended, that is, reversed several times to mix the beads evenly, and the mixing operation should be gentle, not violent vortex oscillation, etc., to avoid antibody denaturation.
3. In the case of immunoprecipitation or purification, it is recommended to set up positive and negative controls.
4. Protein samples should be purified as soon as possible after collection and should always be placed at 4°C or in an ice bath to slow down protein degradation or denaturation.
5. If you use a vacuum pump or other instrument to absorb the supernatant, you must pay attention to the suction strength of the vacuum pump to avoid excessive suction and absorb the collected magnetic beads.
6. When the acidic solution is eluted, the magnetic beads may aggregate, which is a normal phenomenon and does not affect the normal use of the magnetic beads. A 0.1% nonionic detergent (such as Triton X-100, Tween-20, or NP-40) effectively prevents bead aggregation and does not affect the bead's antibody binding efficiency.

7. This product is for scientific research use only.

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